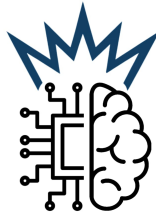


HOME LOCK

A Digital Security Lock System Using Verilog HDL



Project Documentation

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Abstract

The Home Lock project is a digital security system designed using Verilog HDL. It provides controlled access to a protected area through password-based authentication. The system uses a finite state machine to manage password verification, access control, locking, unlocking, and sensor-based operations. The design is functionally verified through simulation to ensure reliable operation under different input conditions.

1. Introduction

Security systems are an important part of modern homes, offices, laboratories, and other protected environments. Traditional mechanical locks require physical keys, which can be lost, duplicated, or misplaced. Digital locking systems provide an alternative by allowing access to be controlled electronically.

The Home Lock system developed in this project is a digital password-based security system implemented using Verilog HDL. Instead of relying on a physical key, the user enters a predefined password. The entered password is compared with the stored password, and access is granted only when the entered password matches the correct password.

The controller is designed as a finite state machine. Each state represents a particular operating condition of the lock. This approach makes the system easier to design, verify, and extend.

The project also considers different situations such as password modification, door sensing, internal locking and unlocking, and access denial. These features make the design more representative of a practical digital home security system.

2. Project Overview

The Home Lock is a digital security system designed using Verilog HDL. It provides secure access to a protected area by verifying the password entered by the user. The system uses an FSM-based control mechanism to manage password verification, access control, locking, unlocking, and sensor operations.

The design is developed as a digital hardware system and verified through simulation to ensure correct operation under different input conditions.

2.1 Objectives

The main objectives of the Home Lock system are:

- To design a digital security lock using Verilog HDL.
- To implement password-based access verification.
- To control locking and unlocking operations.
- To implement an FSM for system control.
- To verify the design through simulation.

3. System Architecture

3.1 Overall Block Diagram

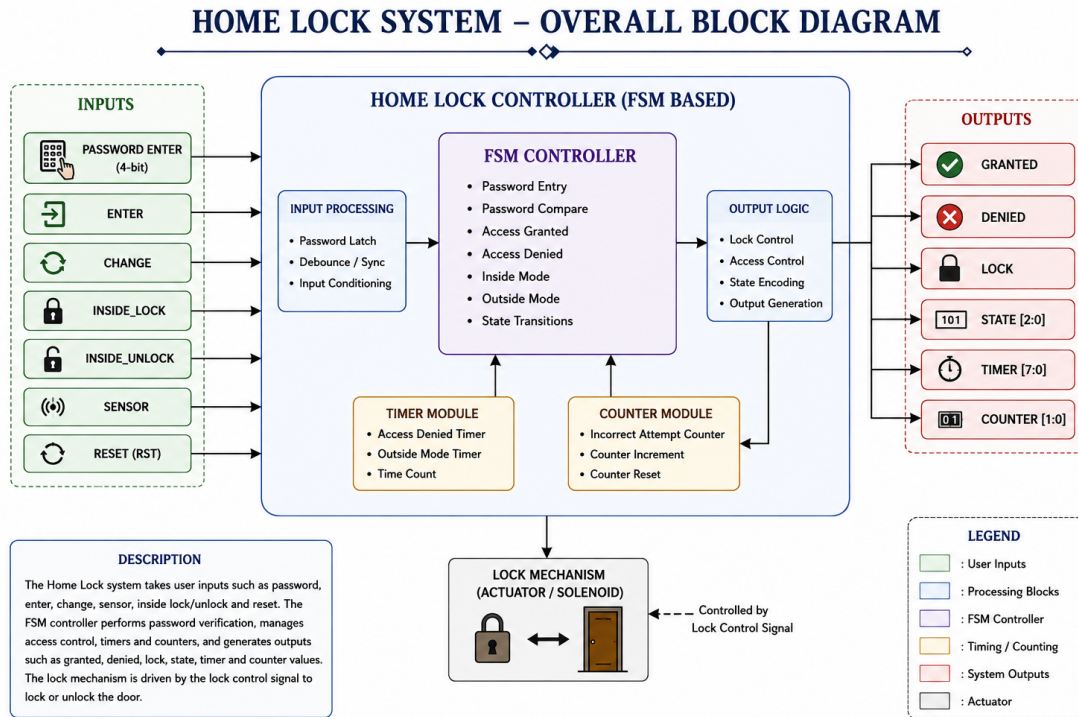


Figure 1: Overall Block Diagram of the Home Lock System

3.2 System Operation

The Home Lock system operates through a sequence of controlled states based on user inputs and sensor conditions.

1. The system is first initialized using the reset signal.
2. The user enters a password and activates the `enter` signal for verification.
3. If an incorrect password is entered repeatedly, the system moves to the access denied state after the allowed attempts.
4. When the correct password is entered, the system moves to the access granted state.
5. When the sensor detects a person, the system changes from access granted state to inside mode.
6. In inside mode, the user can lock or unlock the system using the corresponding control inputs.

7. After unlocking from inside, the system moves to outside mode. The outside mode is controlled using a timer.

4. Design Implementation

The Home Lock is implemented as an FSM-based digital controller using Verilog HDL. The design combines password authentication, access control, sensor monitoring, timing operations, and lock control into a single system.

4.1 Password Entry Module

The password is supplied through the 4-bit `password_enter` input. The user activates the `enter` signal after entering the required value.

The controller uses this input to start the authentication process and move from the password entry stage to the comparison stage.

4.2 Password Verification

The entered password is compared with the predefined password stored in the controller. In this design, the correct password used for verification is `1011`.

A matching password results in successful authentication, while a different value is treated as an incorrect attempt.

4.3 Access Granted and Denied Logic

For a valid password, the controller enters the Access Granted state and activates the `granted` output.

Incorrect passwords are counted as failed attempts. After the specified number of unsuccessful attempts, the controller enters the Access Denied state, where the `denied` output is activated and the corresponding timer operation begins.

4.4 Lock Control Module

The `lock` output represents the current locking condition of the system. Its value is controlled according to the active FSM state and the inputs associated with the locking operation.

The lock can also be controlled from inside using the dedicated lock and unlock inputs.

4.5 Timer and Counter Module

The timer provides clock-cycle based control for time-dependent states. In the Access Denied state, the timer is used to maintain the denied condition for the required duration.

The counter keeps track of password attempts and supports the decision to enter the Access Denied state after repeated failures.

4.6 Inside/Outside Mode

When access is granted and the sensor detects a person, the controller moves into Inside Mode. In this state, `inside_lock` and `inside_unlock` provide direct control over the lock.

After an inside unlock operation, the controller moves to Outside Mode. The sensor status and timer are then used as part of the outside operation.

5. FSM Design

The Home Lock controller is designed using a Finite State Machine (FSM). The FSM controls the sequence of operations based on the password input, enter signal, sensor condition, inside lock/unlock inputs, and timer conditions.

Each state performs a specific function, and the controller changes from one state to another when the required input condition is satisfied. This provides an organized and predictable control flow for the security system.

5.1 State Transition Table

The FSM changes its state according to the current state and the corresponding input condition. The major state transitions of the Home Lock system are shown below.

Table 1: State Transition Table

Current State	Input / Condition	Next State
Password Entry	Enter signal is activated	Compare
Compare	Password is correct	Access Granted
Compare	Password is incorrect	Password Entry
Password Entry	Maximum incorrect attempts reached	Access Denied
Access Granted	Sensor detects a person	Inside Mode
Inside Mode	Inside lock signal is activated	Lock condition
Inside Mode	Inside unlock signal is activated	Outside Mode
Outside Mode	Timer reaches the required value	Password Entry
Access Denied	Denied timer expires	Password Entry

5.2 FSM Flow

The FSM flow starts with password entry and comparison. A correct password grants access, while repeated incorrect attempts lead to Access Denied.

After access is granted, the sensor moves the system to Inside Mode, where lock/unlock inputs control operation. Outside Mode uses a timer to return the system to normal operation.

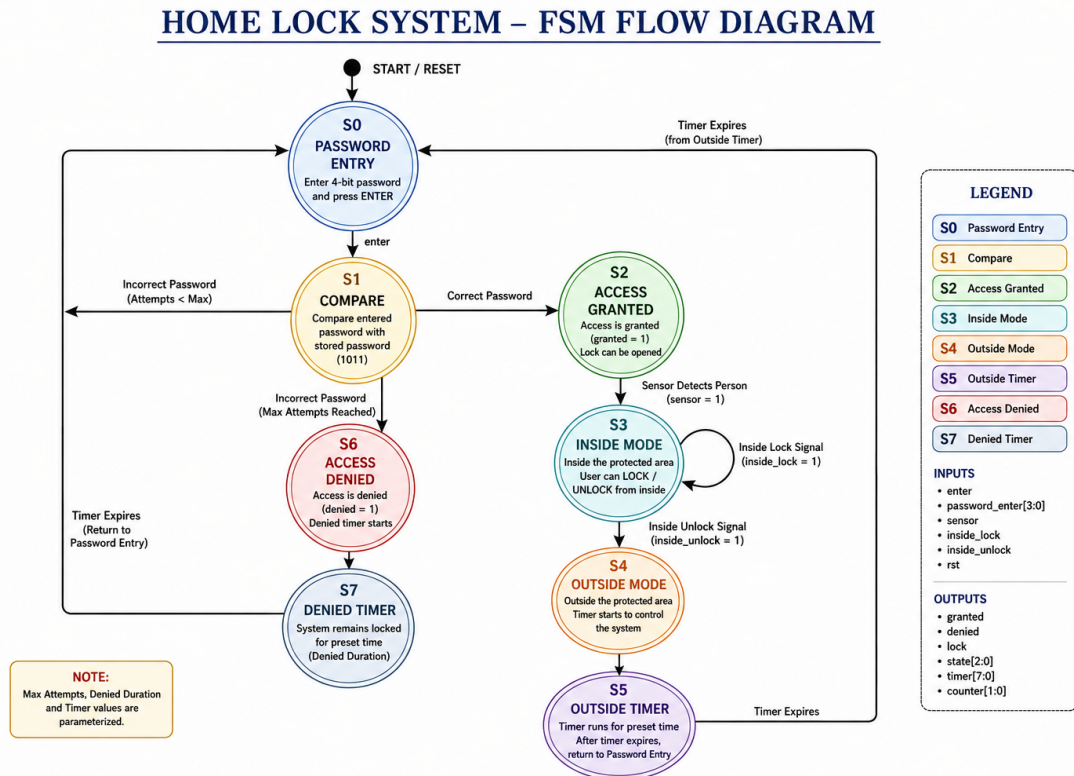


Figure 2: FSM Flow Diagram of the Home Lock System

6. Results

The Home Lock system was functionally verified through RTL simulation. The simulation checks whether the controller responds correctly to different user inputs and state transitions.

6.1 RTL Design

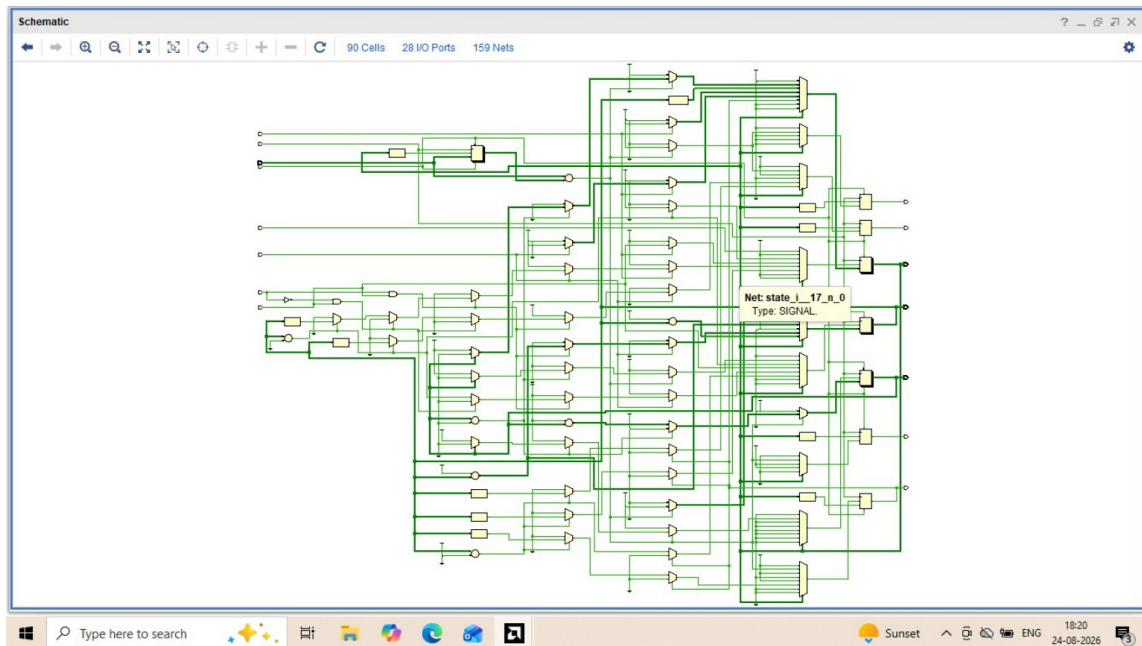


Figure 3: RTL Design of the Home Lock System

6.2 Simulation Results

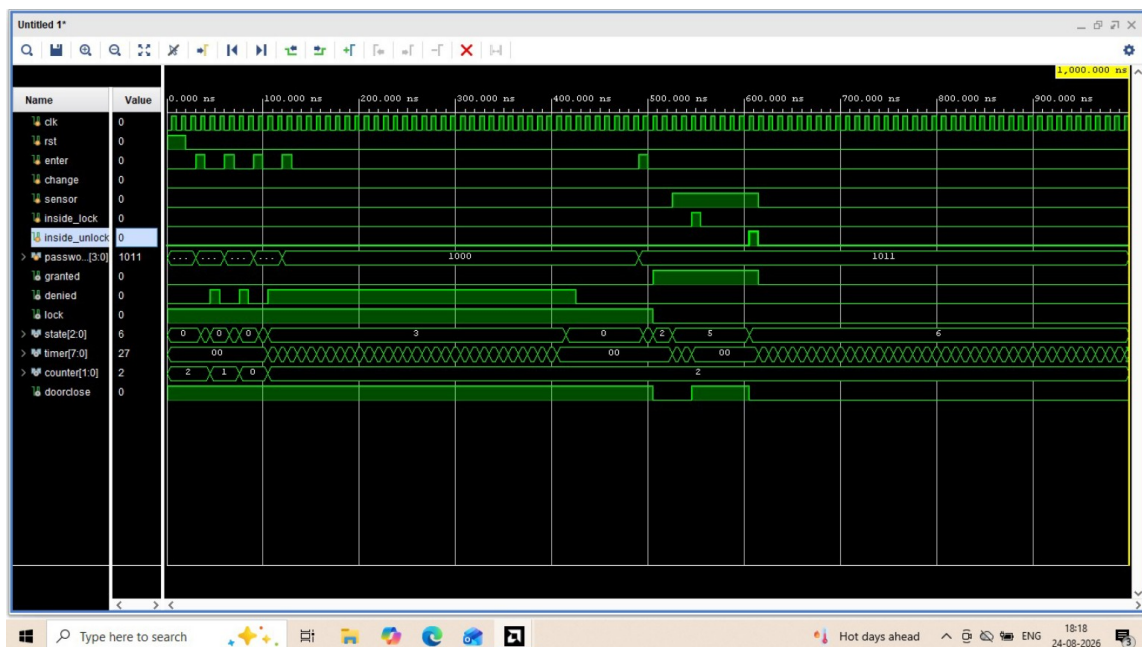


Figure 4: Final Simulation Waveform of the Home Lock System

7. Applications

The Home Lock system can be used as a basic digital security controller in several applications:

- Residential door security systems.
- Office access control.
- Laboratory security.
- Digital cabinets and storage systems.
- Restricted-access rooms.
- Educational FPGA security projects.
- Embedded security control systems.

8. Advantages

The major advantages of the proposed system are:

- Simple digital password-based authentication.
- Fully programmable using Verilog HDL.
- FSM-based structured control.
- Easy to simulate and verify.
- Suitable for FPGA implementation.
- Supports password modification.
- Supports internal and external lock operation.
- Can be extended with additional security features.

9. Conclusion

The Home Lock project successfully implements a password-based digital security system using Verilog HDL. The FSM controls password verification, access control, locking, unlocking, and sensor-based operations.

The design was functionally verified through simulation and showed the expected response for different input conditions. The project demonstrates the practical application of FSMs and digital logic design in a security system.